



CYBERSECURITY • ETHICAL HACKING

# Penetration Testing Report

Footprinting, Reconnaissance & Network Scanning Phases

Target: networkwalks.com

|                            |                                                                                 |
|----------------------------|---------------------------------------------------------------------------------|
| <b>Pentester Name</b>      | Muhammad Ahmad                                                                  |
| <b>Program / Batch</b>     | Cybersecurity & Ethical Hacking Internship                                      |
| <b>Date</b>                | 18 September 2026                                                               |
| <b>Modules Completed</b>   | Footprinting (Kali Linux Tools)<br>Network Scanning (Zenmap / Nmap)             |
| <b>Client / Target</b>     | networkwalks.com (authorized training target)                                   |
| <b>Permission Secured?</b> | Yes                                                                             |
| <b>Phases Covered</b>      | Phase 1: Reconnaissance & Footprinting<br>Phase 2: Scanning & Network Discovery |

## 1. Liability Disclaimer

These activities were performed only against the target and systems for which written permission was already secured, as part of an authorized cybersecurity training exercise. All material in this report is for educational and research purposes only. Nothing in this document should be used to break the law. Unauthorized scanning or access of systems is illegal in most jurisdictions even when no damage occurs, and misuse of these techniques can lead to criminal charges, fines, and other serious consequences. Every action described here is the sole responsibility of the person performing it.

## 2. Introduction

This report documents the footprinting and reconnaissance phase performed against the **networkwalks.com** domain using multiple Kali Linux command-line tools, followed by a full-port network scan performed with **Zenmap** (the graphical front-end for Nmap). Together these two activities demonstrate how a security tester moves from gathering publicly available information about a target to actively probing it to discover live services and open ports.

All footprinting commands were executed from a Kali Linux terminal. The scanning phase was carried out using Zenmap with an "Intense scan, all TCP ports" profile. For every step below, the exact command used, the result observed, a screenshot of the evidence, and a short note on why the finding matters from a security perspective are all included.

## 3. Tools Used

| Tool              | Purpose                                                                            |
|-------------------|------------------------------------------------------------------------------------|
| Kali Linux        | Operating system used to run all reconnaissance commands                           |
| WHOIS             | Retrieve domain registration details (registrar, dates, name servers)              |
| WhatWeb           | Fingerprint web technologies (server software, CMS, plugins, IP address)           |
| Nslookup          | Resolve the domain name to its IP address using DNS                                |
| Curl (-I)         | Read the raw HTTP response headers returned by the web server                      |
| Wafw00f           | Detect whether a Web Application Firewall (WAF) is protecting the site             |
| DNSRecon          | Enumerate DNS records (SOA, NS, MX, TXT/SPF, SRV) and name-server software version |
| Zenmap (Nmap GUI) | Perform an intense, all-port TCP scan to discover live services on the target host |

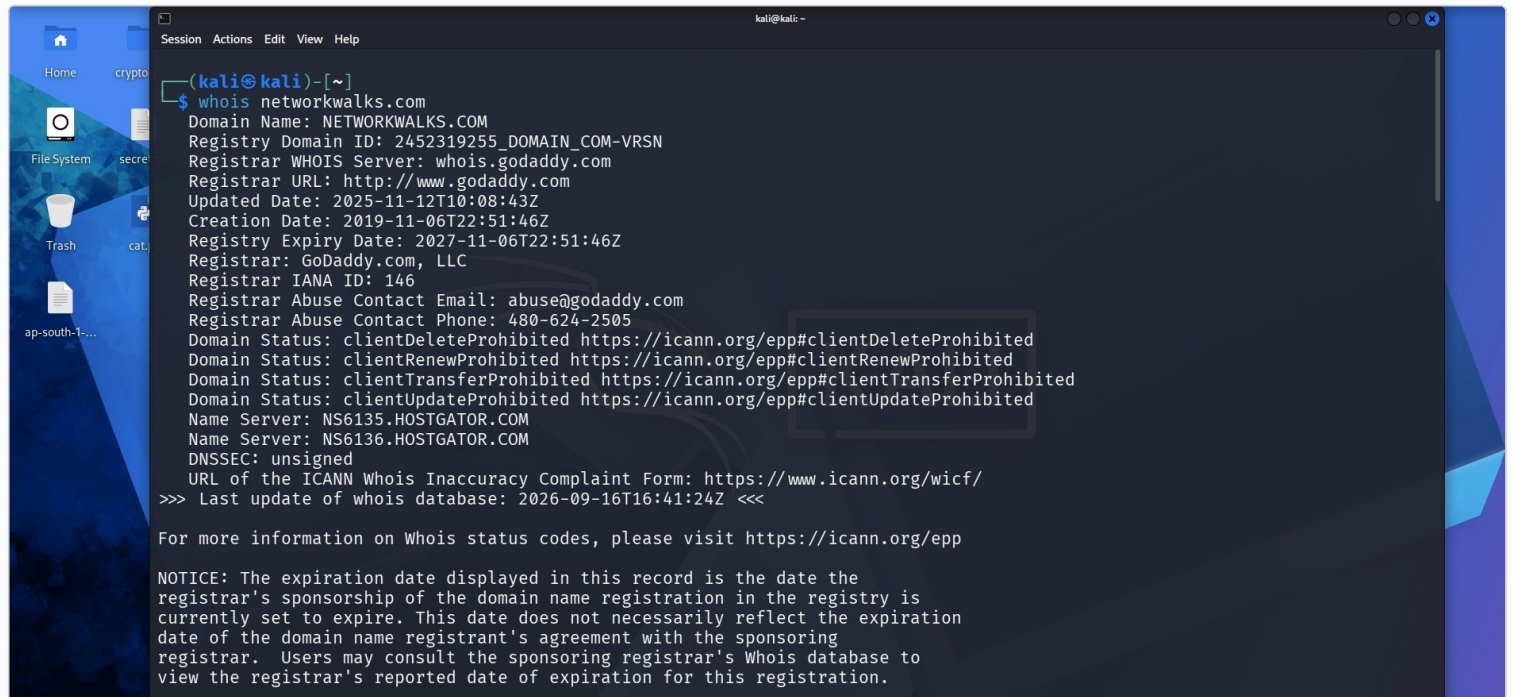
## 4. Activities Performed

### 4.1 Footprinting & Reconnaissance

Reconnaissance was performed against **networkwalks.com** using six Kali Linux tools: **WHOIS**, **WhatWeb**, **Nslookup**, **Curl**, **Wafw00f** and **DNSRecon**. Each tool was used to gather a different category of information about the target, without sending any exploit or attack traffic.

## WHOIS Lookup `$ whois networkwalks.com`

The WHOIS query returned public domain registration details. The domain is registered with **GoDaddy.com, LLC**, was created on **06 November 2019** and is set to expire on **06 November 2027**. The domain uses two Hostgator name servers, **ns6135.hostgator.com** and **ns6136.hostgator.com**, and carries standard registrar-lock statuses (clientDeleteProhibited, clientRenewProhibited, clientTransferProhibited, clientUpdateProhibited). No DNSSEC signing is configured.



```
(kali@kali)-[~]
$ whois networkwalks.com
Domain Name: NETWORKWALKS.COM
Registry Domain ID: 2452319255_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.godaddy.com
Registrar URL: http://www.godaddy.com
Updated Date: 2025-11-12T10:08:43Z
Creation Date: 2019-11-06T22:51:46Z
Registry Expiry Date: 2027-11-06T22:51:46Z
Registrar: GoDaddy.com, LLC
Registrar IANA ID: 146
Registrar Abuse Contact Email: abuse@godaddy.com
Registrar Abuse Contact Phone: 480-624-2505
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Name Server: NS6135.HOSTGATOR.COM
Name Server: NS6136.HOSTGATOR.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-09-16T16:41:24Z <<<

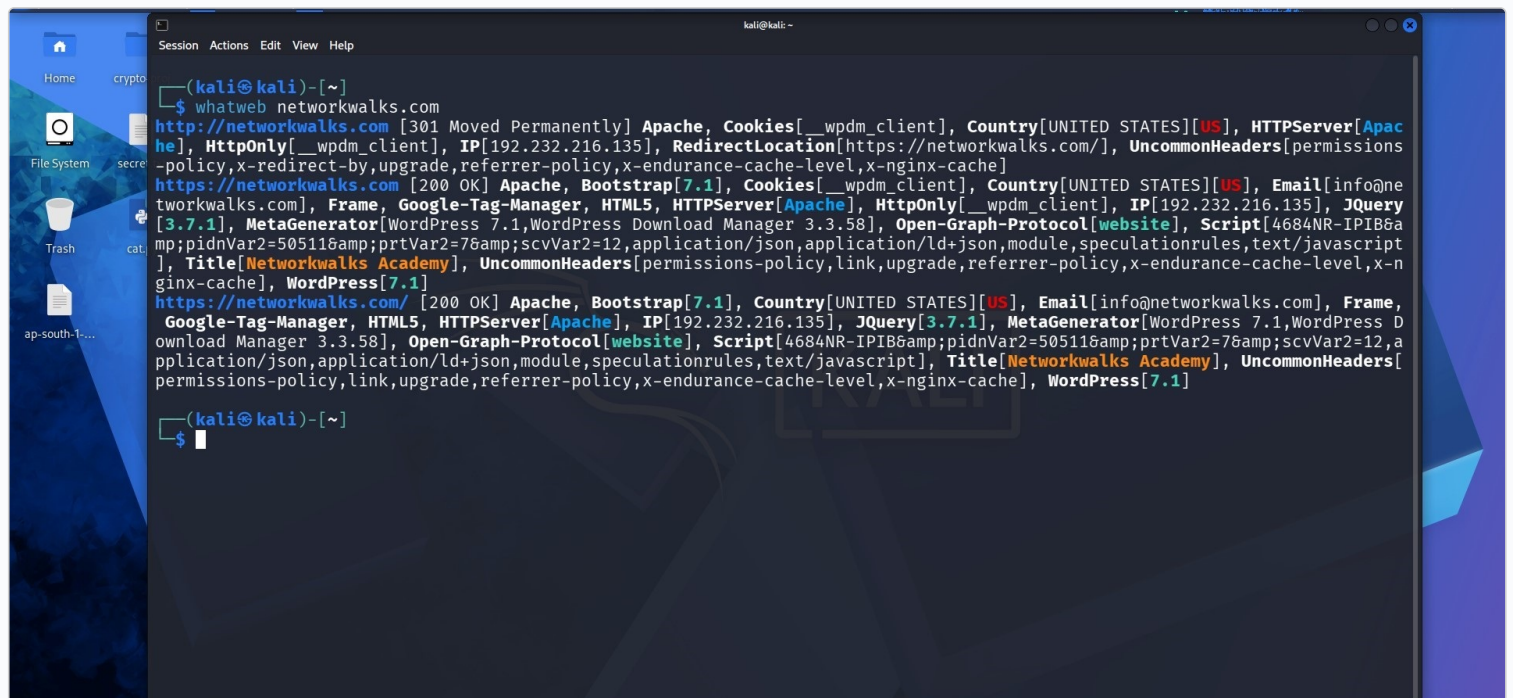
For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar. Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.
```

Figure 1 — WHOIS registration details for networkwalks.com

## WhatWeb Fingerprinting `$ whatweb networkwalks.com`

WhatWeb identified the site as running **WordPress 7.1** with the **WP Download Manager 3.3.58** plugin, served by an **Apache** web server hosted in the **United States** at IP address **192.232.216.135**. The scan also revealed the use of **Bootstrap 7.1** and **jQuery 3.7.1**, a Google Tag Manager integration, and several uncommon response headers (permissions-policy, x-endurance-cache-level, x-nginx-cache), along with the site's HTTP-to-HTTPS redirect behaviour.



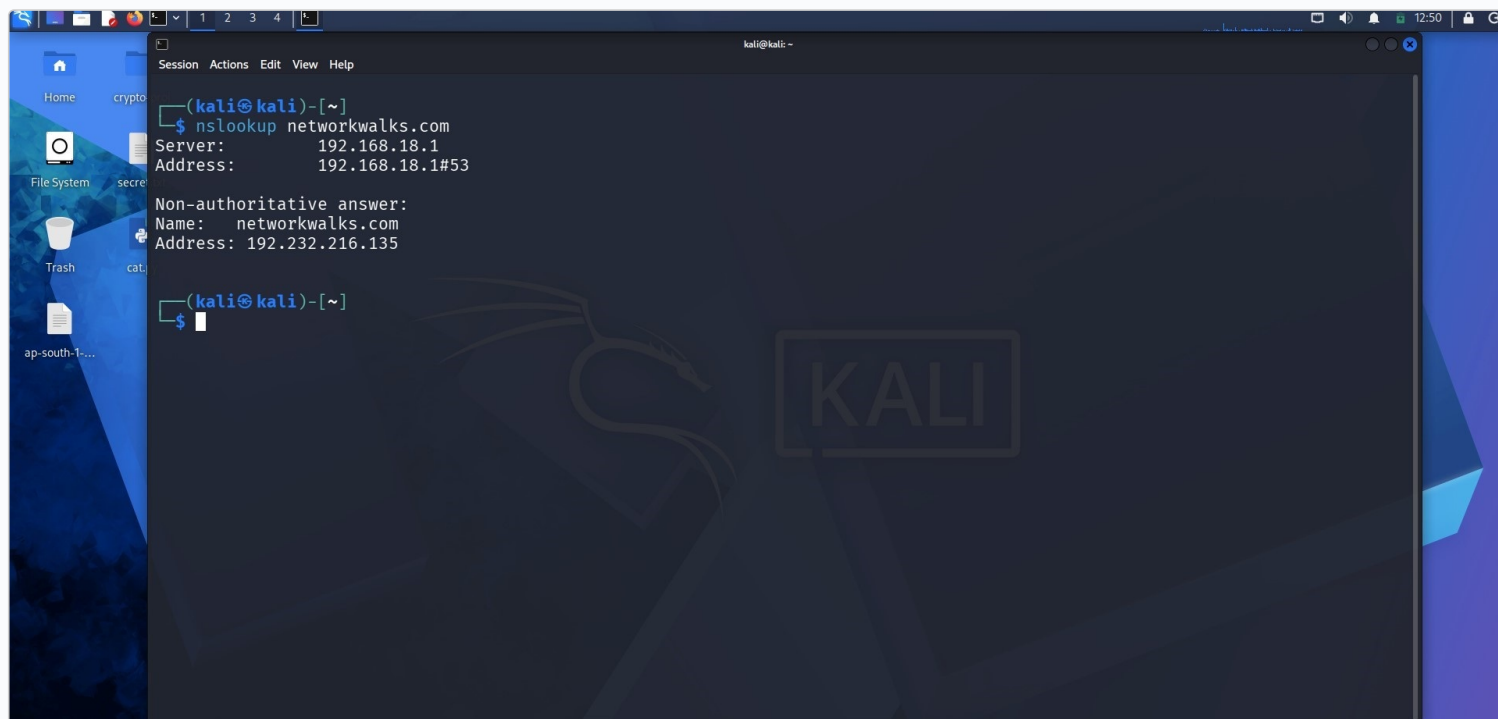
```
(kali@kali)-[~]
$ whatweb networkwalks.com
http://networkwalks.com [301 Moved Permanently] Apache, Cookies[__wpdm_client], Country[UNITED STATES][US], HTTPServer[Apache], HttpOnly[__wpdm_client], IP[192.232.216.135], RedirectLocation[https://networkwalks.com/], UncommonHeaders[permissions-policy,x-redirect-by,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache]
https://networkwalks.com [200 OK] Apache, Bootstrap[7.1], Cookies[__wpdm_client], Country[UNITED STATES][US], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], HttpOnly[__wpdm_client], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1,WordPress Download Manager 3.3.58], Open-Graph-Protocol[website], Script[4684NR-IPIB&amp;pidnVar2=50511&amp;prtVar2=7&amp;scvVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1]
https://networkwalks.com/ [200 OK] Apache, Bootstrap[7.1], Country[UNITED STATES][US], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1,WordPress Download Manager 3.3.58], Open-Graph-Protocol[website], Script[4684NR-IPIB&amp;pidnVar2=50511&amp;prtVar2=7&amp;scvVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1]

(kali@kali)-[~]
$
```

Figure 2 — WhatWeb technology fingerprint of networkwalks.com

## Nslookup DNS Resolution `$ nslookup networkwalks.com`

Nslookup resolved **networkwalks.com** to the IP address **192.232.216.135**, confirming the hosting address identified independently by WhatWeb and later used as the Zenmap scan target.



```
(kali㉿kali)-[~]
$ nslookup networkwalks.com
Server:      192.168.18.1
Address:     192.168.18.1#53

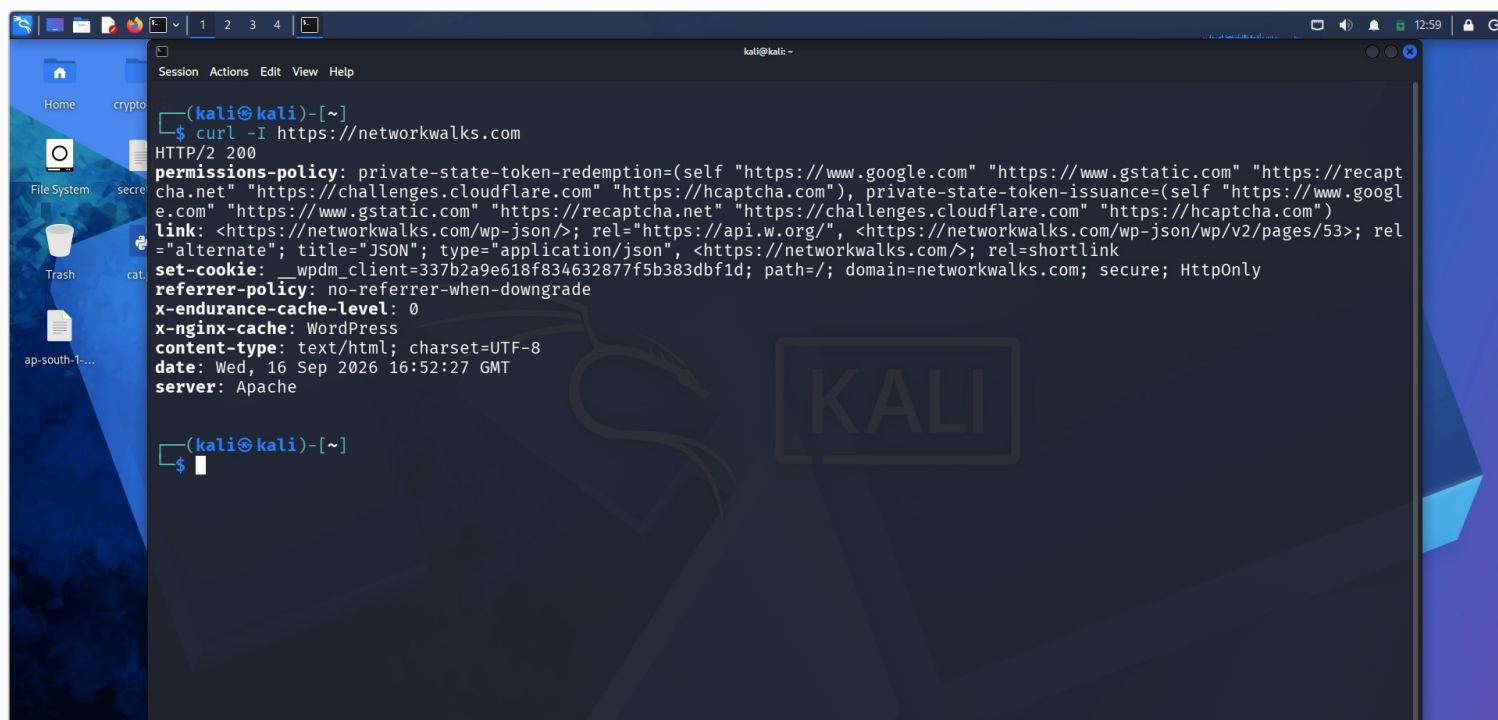
Non-authoritative answer:
Name:   networkwalks.com
Address: 192.232.216.135

(kali㉿kali)-[~]
$
```

Figure 3 — Nslookup resolving networkwalks.com to its IP address

## Curl HTTP Header Inspection `$ curl -I https://networkwalks.com`

The server responded **HTTP/2 200** and returned headers confirming **Apache** as the web server and **WordPress** caching (x-nginx-cache: WordPress). The link header exposed the WordPress REST API endpoint `/wp-json/` and a specific page ID via `wp-json/wp/v2/pages/53`. A permissions-policy header also revealed that the site integrates Google reCAPTCHA and Cloudflare Turnstile challenge services.



```
(kali㉿kali)-[~]
$ curl -I https://networkwalks.com
HTTP/2 200
permissions-policy: private-state-token-redemption=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com"), private-state-token-issuance=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com")
link: <https://networkwalks.com/wp-json/>; rel="https://api.w.org/", <https://networkwalks.com/wp-json/wp/v2/pages/53>; rel="alternate"; title="JSON"; type="application/json", <https://networkwalks.com/>; rel=shortlink
set-cookie: __wpdm_client=337b2a9e618f834632877f5b383dbf1d; path=/; domain=networkwalks.com; secure; HttpOnly
referrer-policy: no-referrer-when-downgrade
x-nginx-cache-level: 0
x-nginx-cache: WordPress
content-type: text/html; charset=UTF-8
date: Wed, 16 Sep 2026 16:52:27 GMT
server: Apache

(kali㉿kali)-[~]
$
```

Figure 4 — HTTP response headers returned by networkwalks.com

## Wafw00f WAF Detection `$ wafw00f networkwalks.com`

Wafw00f identified that the site is protected by **ModSecurity (SpiderLabs)**, a Web Application Firewall, using only 2 requests to make the determination.

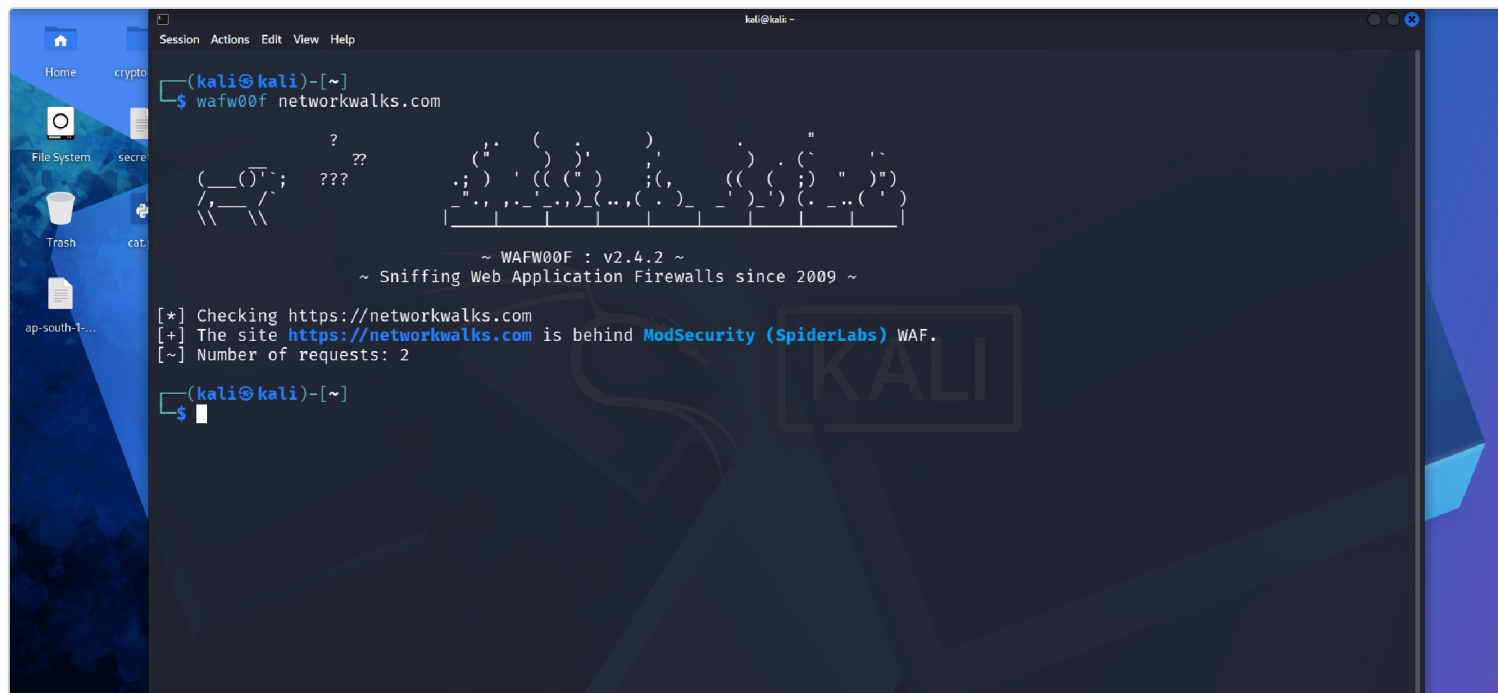


Figure 5 — Wafw00f detecting the ModSecurity WAF in front of networkwalks.com

## DNSRecon Enumeration `$ dnsrecon -d networkwalks.com`

DNSRecon enumerated the domain's DNS infrastructure in detail. It confirmed the SOA/NS records point to **ns6135.hostgator.com** (50.87.144.87) and **ns6136.hostgator.com** (192.232.216.131), and disclosed the exact BIND name-server version, **9.16.23-RH**, for both servers. It also identified the mail exchanger **mail.networkwalks.com**, an SPF TXT record authorizing Hostgator/WebsiteWelcome mail infrastructure, a Google site-verification TXT record, and multiple SRV records for `_autodiscover._tcp` pointing to Cpanel's email autodiscovery service — confirming the site is hosted on a cPanel/WHM shared-hosting platform.

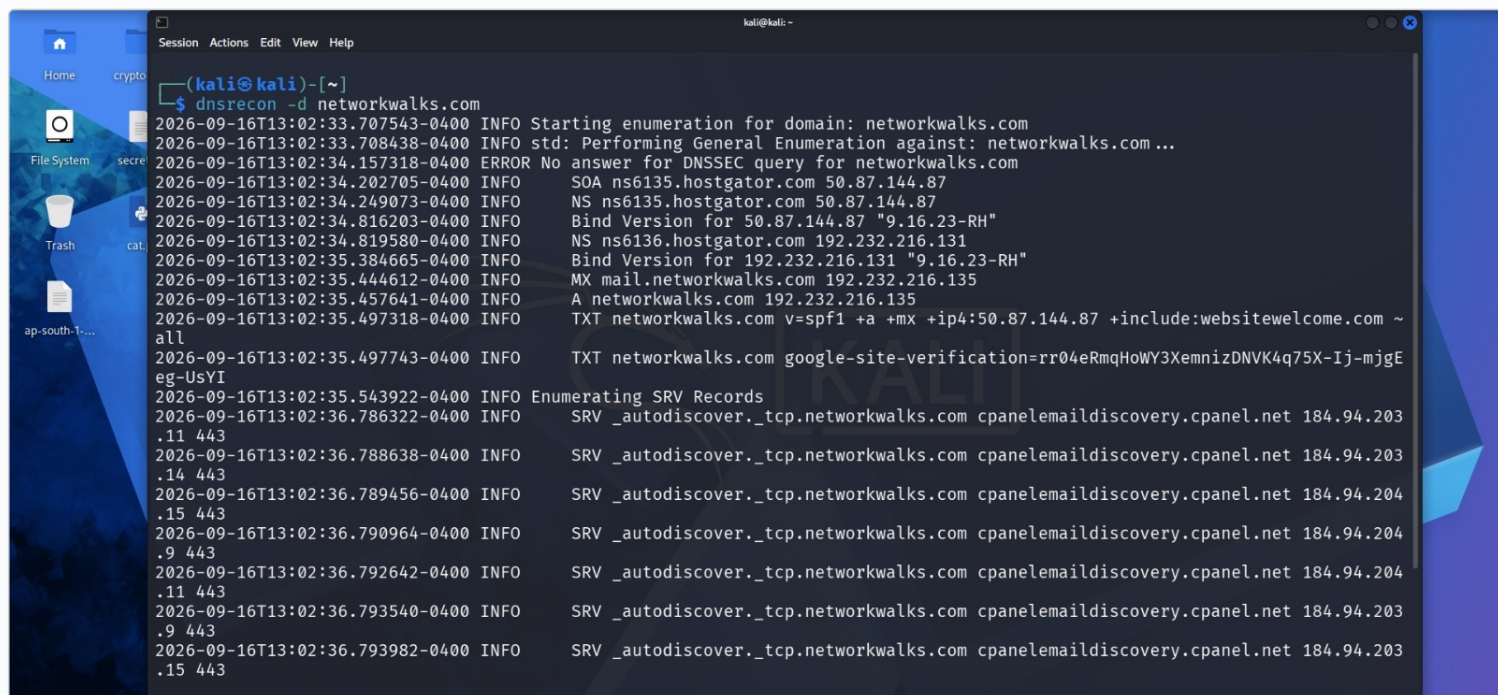


Figure 6 — DNSRecon DNS record enumeration for networkwalks.com

## 4.2 Network Scanning with Zenmap

For the second activity, the resolved IP address **192.232.216.135** was scanned directly using **Zenmap**, targeting the live web server rather than a local LAN, in order to identify every open TCP port and the services running behind them. Full details of the scan configuration,

command, and results are broken out separately in Section 5 below.

## 5. Understanding the Zenmap / Nmap Scan

This section explains the scanning tool on its own, separate from the narrative above, so the methodology and results can be reviewed independently.

### 5.1 What is Zenmap / Nmap?

**Nmap** ("Network Mapper") is the industry-standard open-source tool for network discovery and security auditing. It works by sending specially crafted packets to a target host and analyzing the responses to determine which ports are open, what service is listening on each port, and often the exact software name and version. **Zenmap** is simply the official graphical user interface (GUI) for Nmap: it lets a user pick a pre-built scan "Profile" from a dropdown, automatically builds the equivalent Nmap command line, and then displays the results in organized tabs (Nmap Output, Ports/Hosts, Topology, Host Details, Scans) instead of raw terminal text.

### 5.2 Scan Configuration Used

| Setting                 | Value                                                  |
|-------------------------|--------------------------------------------------------|
| Target                  | 192.232.216.135 (networkwalks.com)                     |
| Zenmap Profile          | Intense scan, all TCP ports                            |
| Equivalent Nmap command | <code>nmap -p 1-65535 -T4 -A -v 192.232.216.135</code> |

Breaking down exactly what each flag in that command tells Nmap to do:

| Flag       | Meaning                                                                                                                                                                                                           |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -p 1-65535 | Scan <b>every</b> possible TCP port (all 65,535), not just the default top-1000, so nothing is missed.                                                                                                            |
| -T4        | "Aggressive" timing template — scans faster by sending probes with shorter delays, appropriate for a stable target/network.                                                                                       |
| -A         | Enables OS detection, service/version detection, script scanning (NSE) and traceroute all at once — this is what allows Nmap to identify not just <i>that</i> a port is open, but what is actually running on it. |
| -v         | Verbose mode — prints progress and discoveries as the scan runs instead of only showing the final summary.                                                                                                        |

Nmap first performs a lightweight **Ping Scan** to confirm the host is alive, then runs a **SYN Stealth Scan** across the full port range. A SYN scan sends a TCP SYN packet to each port and inspects the reply: a SYN/ACK response means the port is **open**, an RST means it is **closed**, and no response (behind a firewall) usually means **filtered**. Because it never completes the full TCP handshake, this technique is fast and relatively quiet compared to a full connect scan.

### 5.3 Ports Discovered

The scan identified the following open TCP ports on 192.232.216.135:

| Port | Likely Service  | Note                                                         |
|------|-----------------|--------------------------------------------------------------|
| 21   | FTP             | File transfer — historically sends credentials unencrypted   |
| 22   | SSH             | Encrypted remote administration access                       |
| 25   | SMTP            | Outbound mail transfer                                       |
| 53   | DNS             | Domain name resolution for the hosted zone                   |
| 80   | HTTP            | Unencrypted web traffic (redirects to HTTPS per Section 4.1) |
| 110  | POP3            | Legacy mail retrieval, unencrypted                           |
| 143  | IMAP            | Legacy mail retrieval, unencrypted                           |
| 443  | HTTPS           | Encrypted web traffic (main WordPress site)                  |
| 587  | SMTP Submission | Authenticated outbound mail submission                       |
| 993  | IMAPS           | Encrypted IMAP mail retrieval                                |
| 995  | POP3S           | Encrypted POP3 mail retrieval                                |
| 3306 | MySQL           | <b>Database port exposed to the public internet</b>          |

```
nmap -p 1-65535 -T4 -A -v 192.232.216.135
```

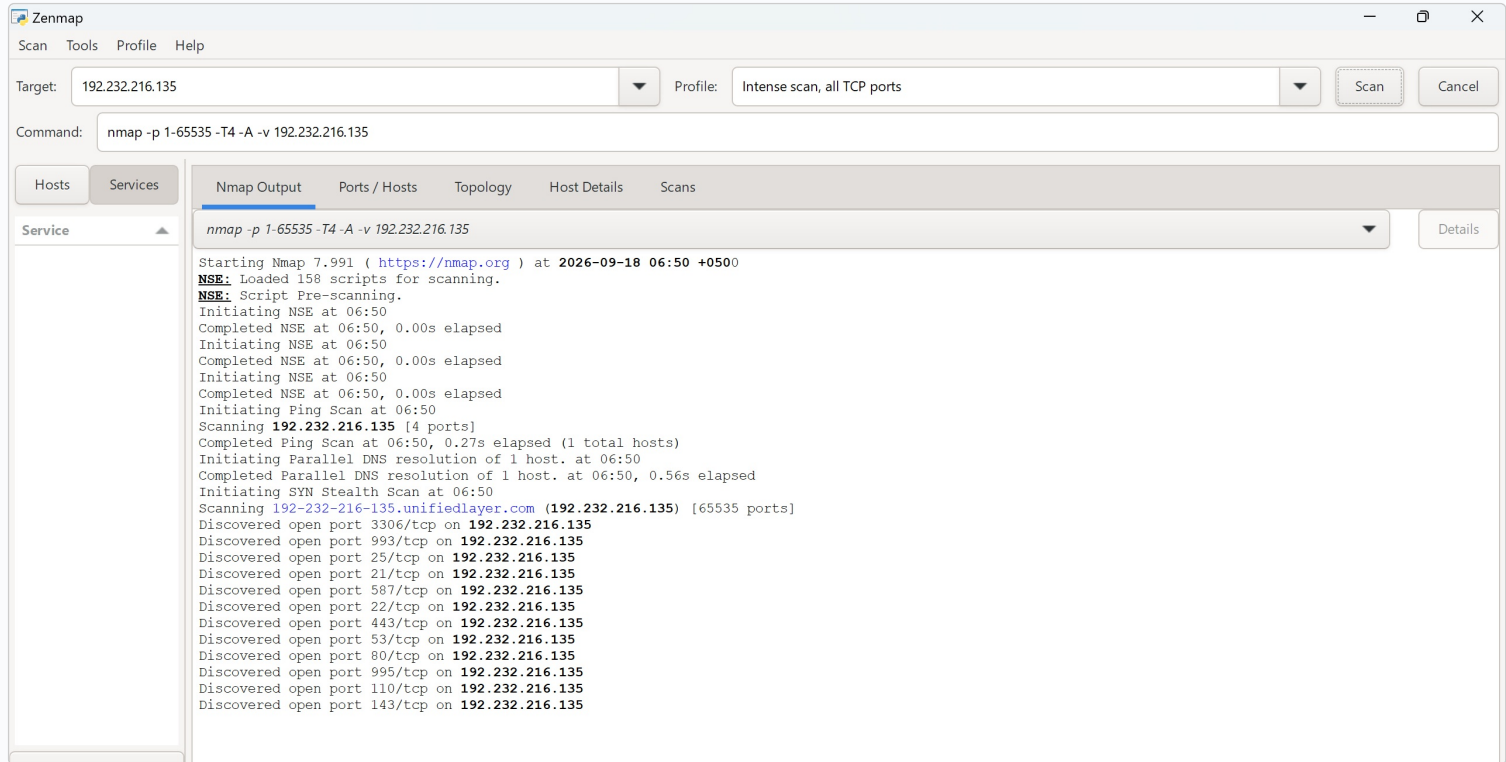


Figure 7 — Zenmap "Intense scan, all TCP ports" output against 192.232.216.135, showing the ping scan, SYN stealth scan, and discovered open ports

## 5.4 Why This Matters

The port list confirms this server is a typical shared-hosting / cPanel-style mail-and-web host: FTP, mail (SMTP/POP3/IMAP and their encrypted equivalents), DNS, and both plaintext and encrypted web ports are all present, which matches the cPanel autodiscovery SRV records found earlier by DNSRecon. The single finding that stands out is **port 3306 (MySQL)** being reachable from the open internet. A database port does not need to be exposed publicly for a WordPress site to function — the application server can reach it over localhost or an internal network instead — so its presence here is worth flagging as a hardening opportunity, independent of whether the database itself is currently misconfigured or vulnerable.

## 6. Risk Analysis / Impact

Based on the footprinting and scanning activities above, the following observations were noted. These are informational findings from reconnaissance, not confirmed, exploited vulnerabilities.

| # | Finding                                                             | Evidence                                                        | Potential Impact                                                                                                           | Risk     |
|---|---------------------------------------------------------------------|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|----------|
| 1 | Web technology & plugin version exposed                             | WhatWeb identified WordPress 7.1 and WP Download Manager 3.3.58 | Attackers can target known CVEs for the exact identified versions                                                          | Medium   |
| 2 | Server IP address identifiable                                      | Nslookup / WhatWeb resolved 192.232.216.135                     | Enables direct scanning of the host, bypassing any CDN in front of the domain                                              | Low      |
| 3 | REST API endpoint & internal page IDs exposed                       | Curl headers revealed /wp-json/ and page ID 53                  | May assist further WordPress REST API enumeration                                                                          | Low      |
| 4 | WAF vendor identifiable                                             | Wafw00f identified ModSecurity (SpiderLabs)                     | Knowing the WAF vendor can help an attacker research known bypass techniques                                               | Low      |
| 5 | Name-server software version disclosed                              | DNSRecon revealed BIND 9.16.23-RH on both name servers          | Exact version can be checked against known BIND vulnerabilities                                                            | Medium   |
| 6 | Mail platform infrastructure exposed                                | DNSRecon SRV records point to cPanel autodiscovery services     | Confirms hosting platform, aiding infrastructure profiling                                                                 | Low      |
| 7 | Full mail service stack reachable (21, 25, 110, 143, 587, 993, 995) | Zenmap intense scan, all TCP ports                              | Larger attack surface; legacy plaintext protocols (FTP, POP3, IMAP) risk credential exposure if used                       | Medium   |
| 8 | Database port (3306/MySQL) open to the internet                     | Zenmap intense scan, all TCP ports                              | Unnecessary direct exposure of the database increases the attack surface for brute-force or exploit attempts against MySQL | Critical |

● Critical ● Medium ● Low

No exploitation or vulnerability validation (e.g. attempting to log into MySQL, brute-forcing FTP, or exploiting a WordPress plugin) was performed as part of this engagement. The presence of a version number, open port, or DNS record does not by itself confirm a system is vulnerable; further authorized testing would be required to validate any of the risks above.

## 7. Recommendations

- Restrict database access** — Firewall port 3306 so MySQL is only reachable from localhost or an internal network, never from the public internet.
- Patch and update WordPress core & plugins** — Keep WordPress and WP Download Manager updated and monitored against current CVE advisories.
- Retire plaintext mail protocols** — Disable or restrict FTP, POP3 (110) and IMAP (143) in favour of their encrypted equivalents (SFTP/FTPS, POP3S, IMAPS) wherever client software allows.
- Patch DNS server software** — Verify the BIND 9.16.23-RH name servers are patched against known CVEs for that version.
- Minimize exposed HTTP header detail** — Review the link, permissions-policy and cache headers to avoid unnecessarily revealing internal page IDs or third-party integrations.
- Keep the WAF tuned and monitored** — Continue running ModSecurity, keep its rule sets updated, and monitor logs for blocked attack attempts.
- Review DNS records periodically** — Confirm only necessary SPF, TXT and SRV records remain published.
- Perform regular authorized external scans** — Re-run full-port scans periodically to catch newly opened or forgotten services.
- Always operate within an authorized scope** — Continue to secure written permission before any reconnaissance or scanning activity.

## 8. Conclusion

This engagement combined passive footprinting with an active, full-port network scan against **networkwalks.com**. The footprinting phase, using WHOIS, WhatWeb, Nslookup, Curl, Wafw00f and DNSRecon, showed how much information about a target's hosting, software stack, DNS infrastructure and security controls can be gathered without ever touching the target directly. The Zenmap scanning phase then confirmed this picture from the network side, mapping every open TCP port on the host and revealing a full mail-and-web hosting stack — including one notable exposure, the publicly reachable MySQL port, that is called out separately as the standout finding of this report.

Together, these two phases illustrate the natural progression of a real reconnaissance effort: build a profile of the target passively first, then verify and extend that profile with direct, low-noise active scanning. All findings here are informational; no exploitation was attempted, and any further testing would require additional written authorization.

